AYOMIDE ABILEWA

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Profile

Works across embedded systems, instrumentation and computer vision. Has built sensing and detection systems from microcontroller nodes up to full-stack applications, and has taught electronics to secondary school students since 2023. Completed an internship on Chevron Nigeria's electrical and instrumentation team.

Education

B.Sc. Electronic and Electrical Engineering 2021–2027 (expected)
Obafemi Awolowo University, Ile-Ife, Osun State, Nigeria

Experience

Facilities Engineering Management Intern (SIWES) Apr–Sep 2026
Chevron Nigeria Limited, Lagos, Nigeria

- Rotated through the Electrical and Instrumentation (E&I) team on field instrumentation and electrical systems.
- Read P&IDs before fieldwork; traced control loop configurations, I/P converters and 3–15 PSI pneumatic signalling.
- On the electrical side: single line diagrams, ESP systems, power transmission, circuit breaker specification.

Research and Development Intern (SIWES) 2024
ACE Quanser Robotics Lab, Obafemi Awolowo University, Ile-Ife, Nigeria

- Operated three Quanser platforms in MATLAB and Simulink: the QDrone quadrotor, the QCar ground vehicle, the QArm 6-DOF manipulator.
- PID control and sensor feedback tuned on live hardware, not in simulation.
- Gave the lab orientation talk on the Quanser Aero 2: architecture, control principles, safety.

Embedded Systems and EEE Tutor 2023–Present
Astar Tutorials, Ile-Ife, Nigeria

- Teach circuit analysis, digital logic and microcontroller programming to engineering undergraduates.
- Sessions are sequenced so the theory arrives just before a student needs it to make a circuit work.

Projects

aniwe Aug 2026
Python, Streamlit, OpenRouter, Vision-language models

- Shipped a convention costume-rating booth in Python and Streamlit: ten hosts, per-session mood and quirk variation, vision-language scoring over seven criteria.
- Text-to-speech falls through three tiers (ElevenLabs, offline Piper, browser speech) and speech-to-text runs locally, so a blocked network costs quality and nothing else.

github.com/ayomide-abilewa/aniwe

AI-Powered Smart Attendance System Feb–Mar 2026

Python, OpenCV, YuNet, SFace

- Owned the vision pipeline and the Flask REST API: YuNet detection, SFace recognition, cosine-similarity matching.
- Built the React and TypeScript frontend over a PostgreSQL schema for students, courses, enrolment, sessions and attendance.
- Defended it at the final-year project review; invited afterwards to build a production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Live Video Pipe Anomaly Detection 2026–Present

Python, YOLOv8, YOLOv11, Weighted Boxes Fusion

- Building a two-stage cascaded detector for real-time pipe anomaly identification in industrial inspection.
- Two YOLO models run in parallel, one on RGB and one on edge-enhanced input, merged with Weighted Boxes Fusion to hold precision across lighting and surface conditions.

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded and Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI and Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation and Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Electrical Systems: SLD interpretation, Circuit breaker sizing, ESP systems, Power transmission, Arc flash safety, Hazardous area classification

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

Leadership and Teaching

SPAW Project Lead and Syllabus Team Lead 2024–Present

IEEE OAU Student Branch

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.
- Design all lesson plans, tutor guides and laboratory exercises for the programme.
- Session packs go out as PDFs with circuit diagrams, code and a troubleshooting section.

Core Instructor 2023–Present

Circuit Zero to Hero (ZTH)

- Run practical electronics workshops for students who have never built a circuit.

Training and Certifications

Operational Excellence Training, Chevron 2026

Cyber Security and Data Privacy Awareness, Chevron 2026

Business Conduct, Ethics and Conflict of Interest, Chevron 2026

FPGA Design and HDL/Verilog Workshop (SWEP), Faculty of Engineering, Obafemi Awolowo University 2024

Robotics Lab Orientation and Operations, ACE Quanser Lab, Obafemi Awolowo University 2024